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IEEE Guide for Dissolved Gas Analysis in Transformer Load Tap Changers

IEEE Power and Energy Society

Sponsored by the
Transformers Committee

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3 Park Avenue
New York, NY 10016-5997
USA

IEEE Std C57.139™-2015
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Approved 5 December 2015

IEEE-SA Standards Board

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Abstract: General types of load tap changer (LTC) mechanisms, breathing configurations, and electrical designs, described for reference for determining methods of testing and evaluating dissolved gases in mineral-based transformer oils in LTCs, are discussed and recommended in this guide. Measurements of dissolved gas concentrations in the LTC oil are required to implement this guide. This guide is not manufacturer specific, rather category specific.

Keywords: DGA, dissolved gas-in-oil, IEEE C57.139™, load tap changer, LTC

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Paul Boman
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Mark Perkins
John Pruente
Scott Reed
Oleg Roizman
Richard Simonelli
Thomas Spitzer
Markus Stank
Bengt-Olof Stenestam
Craig Stiegemeier
Pontus Sundqvist
Mark Tostrud
Jane Ann Verner
Dharam Vir
Melvin Wright
Shuzhen Xu
Dong Zhan

Major contributions were received from the following individuals:

James Dukarm
Michel Duval
Rainer Frotscher
Fredi Jakob
John Pruente

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Introduction

This introduction is not part of IEEE Std C57.139-2015, IEEE Guide for Dissolved Gas Analysis in Transformer Load Tap Changers.

This guide is a revision of the first issue of a document to assist the industry in the understanding and interpretation of dissolved gas analysis (DGA) in LTCs. It makes use of statistical tools to generate norms for fault gas levels and fault gas concentrations to facilitate this understanding. The guide also provides information on LTC classification schemes, result codes, and analytical tools to assist the user with LTC DGA interpretation.

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1. Overview

1.1 Scope

This guide discusses and recommends methods of testing and evaluating dissolved gases in mineral based transformer oils found in *load tap changers* (LTCs). General types of LTC mechanisms, breathing configurations, and electrical design will be included for evaluation criteria in determining when mechanical damage or failure has occurred. *Dissolved gas analysis* (DGA) of the oil in the LTC is required. This guide is not manufacturer specific, rather category specific.

1.2 Purpose

The purpose of this guide is to assist the responsible parties who are in charge of the operation and maintenance decisions in evaluating the condition of an LTC without the need to de-energize the transformer to inspect the LTC in question. Additionally, repairs to the LTC can be made in a timely fashion based on accurate interpretation of the gas analysis, minimizing premature repairs or post failure rebuilds.

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1.3 Limitations

Only suitably trained personnel following the ASTM D923 procedure should perform oil DGA sampling.[1] The measurement of gas concentrations should be performed by an analytical laboratory according to the methods defined in ASTM D3612, or by properly trained personnel using a portable instrument, or by means of an on-line monitoring device (see Soundarrajan [B9]).[2]

A properly trained and experienced person should perform the interpretation of DGA data. Computer assistance including utilization of spreadsheets, databases, analytical software, graphical methods (e.g., Duval triangle), maintenance management systems, etc., is useful. Additional information about the LTC and the transformer on which it is mounted, including operation, maintenance, test, and environmental history, may be needed for a detailed interpretation.

For various reasons, some of which are explained in this guide, DGA results can sometimes be false or misleading, and sometimes DGA can fail to identify an actual fault that may be present. As with any diagnostic tool, care must be used when analyzing DGA results. Unexplained variations or changes in the gas concentrations can lead to inaccurate conclusions. Results should be confirmed by additional DGA testing. Diagnostic and repair decisions should not be made based on DGA results only, particularly not on a single DGA sample. In addition, other tests and consultation with experts should be employed where possible before inspection and/or repairs are performed on the equipment. Local operating conditions, requirements, and safety issues should be considered.

1.4 Safety warning

The following safety warning applies only to transformer LTCs to be tested in the field, not to factory testing. Refer to factory test codes for safety warnings for these situations.

WARNING

Combustible gases dissolved in the oil in LTCs can reach concentrations that can create a flammable mixture in air exposed to the oil. This oil should be handled as though it might create a flammable mixture in air or tested to determine if it does contain amounts of combustible gases that can form a flammable mixture in air.

Oil with gas levels exceeding 1000 µL/L of dissolved hydrogen and/or exceeding 10 000 µL/L of total dissolved combustible gas may create a flammable gas mixture when exposed to air and should be evaluated. The flammability is dependent on the solubility of the gases in oil and their lower explosion limits that are gas specific, so these values only provide a guideline to seek further information.

Prudence should be exercised in handling oil containing flammable gas concentrations at or above these limits.

See Pereira [B8] for additional information.

2. Normative references

The following referenced documents are indispensable for the application of this document (i.e., they must be understood and used, so each referenced document is cited in text and its relationship to this document is explained). For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments or corrigenda) applies.

[1] Information on references can be found in Clause 2.
[2] The numbers in brackets correspond to those of the bibliography in Annex F.

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ASTM D3612, Standard Test Method for Analysis of Gases Dissolved in Electrical Insulating Oil by Gas Chromatography.[3]

ASTM D923, Standard Practices for Sampling Electrical Insulating Liquids.

3. Definitions

For the purposes of this document, the following terms and definitions apply. The *IEEE Standards Dictionary Online* should be consulted for terms not defined in this clause.[4]

arcing gases: Hydrogen and acetylene gases may be referred to as *arcing gases*. Hydrogen is the primary gas generated during low-intensity discharges. Acetylene is the primary gas generated during high-intensity arcing faults in the 700 ˚C to 1800 ˚C temperature range.

coking: The formation of a hard carbonized deposit on the contacts of a tap changer. This process occurs due to the breakdown of oil from heat and arcing across the tap changer contacts that are immersed in oil.

continuous on-line filtration: Removal of oil insoluble contaminants by use of a permanently installed on-line filter and continuous circulation of the oil.

desiccant breather unit: An oil-filled free-breathing load tap changer (LTC) compartment with the addition of a dehydrating breather to remove moisture from the air that is in contact with the oil. Typically, the internal pressure inside a *desiccant breather* is slightly higher than atmospheric.

dissolved gas analysis (DGA): The measurement and interpretation of dissolved gas concentrations in the insulating liquid of electrical equipment. The interpretation can identify certain kinds of faults, sometimes in very early stages, and provide a rough indication of severity.

free breather unit: An oil-filled load tap changer (LTC) compartment mounted on the transformer in which the air space above the tap changer oil freely exchanges gases with the outside atmosphere.

heating gases: Methane, ethane, or ethylene gases are referred to as *heating gases* since they are generated when insulating liquid is exposed to higher than normal operating temperatures generated by hot metallic surfaces in the 150 ˚C to 500 ˚C temperature range.

non–invasive method: Any method of testing a load tap changer (LTC) that does not alter the properties of the oil, components, or insulating materials. Such methods may include, but are not limited to, DGA, fluid quality, thermography, and ultrasonics.

sealed tank unit: An oil-filled load tap changer (LTC) compartment that is totally sealed to the atmosphere that vents upon reaching a predefined pressure setting. Typically the compartment contains a one-way valve. This arrangement can result in development of a partial vacuum in the compartment when the oil temperature drops.

total dissolved heating gas (TDHG): The sum of the concentrations of heating gases methane, ethane, and ethylene, in μL/L (ppm_v), in the oil.

[3] ASTM publications are available from the American Society for Testing and Materials (http://www.astm.org/).
[4] The *IEEE Standards Dictionary: Glossary of Terms & Definitions* is available at http://dictionary.ieee.org/.

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4. Nature, purpose, and basis for DGA for LTCs

4.1 Nature of LTC DGA

Dissolved gas analysis (DGA) for LTCs is a process of measurement, identification, and interpretation of gases dissolved in the LTC oil. Usually at regular intervals, a small sample of oil is drawn according to ASTM D923 (see DGA sampling), and the sample is sent to an analytical laboratory, which measures the dissolved gas concentrations with a gas chromatograph according to ASTM D3612 and sends back a report. Alternatively, the analysis may be done on-site with a portable gas analyzer instead of sending the sample to a laboratory. An on-line dissolved gas monitor attached to an LTC is another possible source of DGA data (see Soundarrajan [B9]). The process of sample collection, transportation, and analysis of its gas content is subject to various hazards, discussed as follows, that can affect the quality and usefulness of the data produced.

Provided that data quality is good, the current and historical DGA data for an LTC, together with supporting information about operating conditions, maintenance activities, and so on, can be interpreted to determine whether or not there is evidence of a fault. If a fault is suspected, it may be possible to assess its general nature and relative severity. Usually an apparent indication of a fault by DGA needs to be confirmed by resampling and possibly by supplementary tests.

4.2 Purpose of LTC DGA

DGA provides an effective and economical means of detecting problems or confirming the apparent absence of problems. The overall purpose of LTC DGA is to improve safety and reliability while reducing cost. Cost reduction is achieved by avoidance or mitigation of outages, equipment damage, failures, and by optimization of operation and maintenance. Safety and reliability are improved through awareness of equipment condition, early detection of faults, avoidance of power quality concerns, and monitoring of suspect equipment.

4.3 Gas formation, retention, and dissipation

Because of the fact that LTCs contain switching contacts immersed in the insulating oil, it is expected that combustible gases—especially hydrogen (H_2), methane (CH_4), ethane (C_2H_6), ethylene (C_2H_4), and acetylene (C_2H_2)—are usually formed as a byproduct of normal operation. Arcing in oil produces mainly acetylene and hydrogen, with minor amounts of the heating gases methane, ethane, and ethylene. The rate and quantity of combustible gas formation depend greatly on the design of the LTC, load current, frequency of tap changing, and environmental conditions. The combustible gas formed in an LTC may gradually dissipate into the atmosphere if the LTC has a free or desiccant-type breathing system, or the gas may accumulate and then occasionally vent into the atmosphere through a pressure relief valve in a sealed tank system. Fortunately, in spite of the variability of gas accumulation even under normal conditions in non-faulty equipment, it is possible to discern certain patterns that can be used to distinguish between normal and faulty behavior in many cases.

In vacuum interrupter type LTCs, where the current switching is confined in interrupter vacuum bottles, the operation of reversing or bypass switches and selector contacts immersed in the insulating oil characteristically forms very little, if any, gas. In non-vacuum LTCs with separate oil compartments for the selector contacts and the diverter contacts, the dissolved combustible gas concentrations are generally much lower in the selector compartment than in the diverter compartment. In non-vacuum LTCs, normal dissolved combustible gas formation consists mainly of arcing gas and smaller amounts of heating gases. *Resistor* type LTCs usually contain higher amounts of heating gases than *reactor* type LTCs due to the

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heating of the transition resistors during a tap-changing operation. The gas concentrations depend on the LTC model, the actual transformer load, the winding capacities, the frequency of tap-changing operations, and the breathing configuration.

Typical and unusual gas concentrations depend on the design of the LTC and its operating conditions. Typical gas levels in an arc furnace transformer LTC would be very high compared to the typical levels for an urban distribution substation transformer LTC of the same type. A single-compartment LTC with oil-immersed interrupter contacts would have much higher typical gas levels than a vacuum interrupter type LTC.

In general, for any particular type and application of LTC, the quantity of combustible gas found in an LTC oil compartment is proportional to the amount of tap-changing activity. Unusually high (for the LTC type) combustible gas concentrations may indicate an unusual, possibly fault-related, amount of switching or some other kind of fault. In vacuum interrupter type LTCs, high combustible gas concentrations may indicate faulty vacuum interrupters, mechanical deficiencies, or problems with the oil-immersed bypass and reversing switches. There is also value in sampling an LTC that has been disabled from operating/performing a tap change but remains in service electrically where contacts remain in the circuit. Contact pressures and resistance can change over time without operation and can be detected by DGA sampling. Conditions may include contact coking and overheating.

Aside from the gas concentrations, ratios such as ethylene/acetylene, total dissolved heating gas (TDHG)/acetylene, ethane/methane, methane/acetylene and ethylene/ethane have been found to be useful indicators of relative rates of formation of gases. These ratios have been defined so that an increase in their value indicates an active fault. If other ratios are used, it is recommended that they be set up so that fault-related gas production increases the ratio value. Unusual gas ratio values can also indicate the presence of faults, sometimes even when gas concentrations are entirely normal.

LTC DGA is complicated by the loss of combustible gas from the oil due to air breathing or other exposure to air due to leaky on-line filtration systems. While gas concentrations may be reduced gradually by diffusion to the atmosphere, or more rapidly by occasional pressure relief venting, typically the most useful gas ratios are not greatly affected. Gas concentrations in LTC oil compartments with closed or sealed breathing systems can accumulate to extremely high levels, even in the absence of faults, but fault-related gas production can often still be detected by unusual gas ratio values.

4.4 Basis of LTC DGA

The basic principles upon which LTC DGA rests are as follows.

Each non-faulty tap-changing operation involving oil-immersed contacts, depending on the LTC type and operating conditions, generates characteristic amounts of acetylene, hydrogen, and other combustible gases, which dissolve in the oil and persist for some time, depending on the LTC breathing configuration. Higher load and/or increased tap-changing operations can result in higher dissolved combustible gas concentrations. Similarly, the fact that in non-faulty LTCs the relative rates of production of the various combustible gases tend to be fairly stable means that the ratios of those gases tend to fall within a certain range of values, depending on the LTC type.

Faults and some other unusual conditions tend to change the usual pattern of combustible gas formation. Some kinds of faults, and unusually high tap-changing frequency (which may or may not be fault related), tend to generate abnormally high amounts of combustible gas. Some faults, especially those caused by coking or deterioration of oil-immersed arcing contacts, tend to increase the rate of production of heating gas relative to arcing gas, and consequently the corresponding gas ratios take on values outside their historic range. Experience has shown that some faults can produce high gas concentrations without necessarily altering the gas ratios very much, while other faults can produce abnormal gas ratios without necessarily producing unusually high levels of gas.

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There are standard and well-understood statistical methods for distinguishing between ordinary behavior, which produces a characteristic distribution of values of a variable such as a gas concentration or a gas ratio, and exceptional behavior, which tends to produce extreme values of that variable not fitting the usual pattern. Limit values of the most useful gas concentrations and gas ratios can be defined by statistical analysis of the DGA data for a population of similar LTCs for separating ordinary values from atypical ones. Sample procedures for deriving DGA limits from DGA data are described in Annex B.

In populations of vacuum interrupter type LTCs where the normal range of values of gas concentrations, especially acetylene, is very low, the gas ratios may be redundant for fault detection, and it may even be impractical to derive limits for the main gas ratios.

5. Norms for LTC DGA

5.1 LTC DGA variables

Determination of which measured or derived quantities are most useful for detection and assessment of faults in LTC DGA presents a specific challenge. Part of the question is which gas concentrations and ratios are the most useful to discriminate between normal and faulty conditions, and another part of the question is a matter of statistical robustness.

For some LTC types, the use of ratios using hydrogen should be avoided due to its low solubility. In addition, because of its low molecular weight and low solubility in oil, hydrogen is easily lost by diffusion out of the oil and escapes to the atmosphere through the LTC breather, defective syringes, sample mishandling, or air exposure during sampling or analysis.

The acetylene (C_2H_2) concentration is related to arcing in oil. It is diagnostically useful and is also important in ratios. Acetylene is always present in the oil of non-vacuum LTCs and usually present in operating vacuum type LTCs in small quantities near or below the method detection limit. Vacuum type LTCs may generate acetylene during tap changes due to slight arcing between moving and fixed bypass contact switch operations while opening and closing the bypass circuit.

Ethylene (C_2H_4) is one of the heating gases. It is produced when oil is heated to approximately 350 °C. Deteriorated or coked arcing contacts tend to be associated with abnormal production of ethylene. The other heating gases, methane (CH_4) and ethane (C_2H_6), tend to be produced at lower temperatures than ethylene and are typically produced by resistor heating in a resistance type LTC. The concentrations of all these gases are potentially useful indicators of unusual thermal conditions.

Total dissolved heating gas (TDHG), which is the sum of the concentrations of methane, ethane, and ethylene, is also useful as an indicator of unusual thermal conditions. Like the individual heating gas concentrations, TDHG can be used in ratios.

5.2 Types of limits

For a population of similar LTCs subject to similar operating conditions, a set of norms (limits) should be determined for the purpose of discriminating between typical values of the DGA variables and atypical values that are likely to be associated with faults. If there are separate LTC oil compartments for arcing vs. non-arcing oil-immersed switches, each compartment should have its own set of norms corresponding to the pattern of gas formation expected in that compartment and the kinds of faults that may occur in it. Given sufficient data, the process of deriving appropriate norms is a statistical calculation, which is

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described in Annex B. If appropriate data are not available, it may be useful to make tentative use of norms derived by others for the same type, or a similar type, of LTC.

5.2.1 Minimum reliable value for gas concentrations

For individual gas concentrations and TDHG, it is advisable to set limits specifying how large they must be for reliable interpretation and for calculating ratios. Typically, good measurements of gas concentrations by a lab or portable gas analyzer have an uncertainty of about plus or minus the greater of 1 μL/L (ppm_v) (the method detection limit) or 10% of the measurement value. For example, a reported value of 4 μL/L (ppm_v) would represent 4 ± 1 μL/L (ppm_v), which has a relative uncertainty of ±¼, i.e., ±25%. A reported value of 10 μL/L (ppm_v) or more would have a relative uncertainty of ±10%.

NOTE—The relative uncertainty of reported gas concentrations can be higher than 10% if there are irregularities with sampling, sample handling and transportation, or laboratory processes.[5]

A suggested minimum reliable value for any individual gas concentration that is to be used for LTC DGA or for calculating a gas ratio is 10 μL/L (ppm_v). An appropriate minimum reliable value for TDHG would be 10 to 15 μL/L (ppm_v).

5.2.2 Caution limits for gas concentrations

To identify cases where a gas concentration (or TDHG) is high enough that abnormal switching activity or a fault may be suspected, a caution limit should be set. Appropriate caution limits depend on the LTC model and the operating conditions. If limits generated by another LTC population are adopted in lieu of limits derived from the user's own LTC population's DGA data, those limits must be treated with extra suspicion and adjusted in light of actual experience.

A statistical method for deriving a caution limit for an individual gas or TDHG is described in Annex B. Annex C presents an example of the application of limits obtained using this method. Statistical methods applied to TDHG are more likely to succeed in producing an appropriate limit because TDHG concentrations usually have a well-behaved statistical distribution. Conversely, individual gas concentrations seen in LTC data generally have irregular distributions, which make producing an appropriate statistical limit less likely.

It is suggested that caution limits be provided for TDHG and acetylene. If it is possible, statistically or through extensive familiarity with the LTCs, to assign other caution limits, they should be assigned to ethylene, methane, and ethane in descending order of usefulness.

5.2.3 Caution and warning limits for gas ratios

Experience with non-vacuum LTC DGA has shown that in non-faulty LTCs the combustible gas ratios, especially those that compare any heating gas or combination thereof with acetylene, have values that are relatively independent of the number and timing of operations and gas loss to the atmosphere. Faults tend to change those ratios markedly, so it is possible to detect many LTC faults by looking for atypical values of gas ratios. The determination of what is or is not an atypical value of a quantity is a well-understood statistical problem. In some vacuum type LTCs where the caution limits for heating gas and acetylene concentrations may be very low, gas ratios may be less useful.

[5] Notes in text, tables, and figures of a standard are given for information only and do not contain requirements needed to implement this standard.

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For most non-vacuum LTCs, the ethylene/acetylene ratio is a very useful indicator of the condition of the arcing contacts. The TDHG/acetylene ratio can also be used for the same purpose. For many LTC types, the ethylene/ethane and methane/acetylene ratios seem to complement the ethylene/acetylene and TDHG/acetylene ratios by indicating incipient faults to which the latter ratios are slow to respond.

For ease and consistency of interpretation, all combustible gas ratios used for LTC DGA should be defined so that their values get worse by increasing. For example, the ethylene/acetylene ratio is preferred to the acetylene/ethylene ratio.

The statistical procedure described in Annex B is used to derive Caution (2) and Warning (3) limits for gas ratios. See Table 1. The derived limits represent critical values for evaluating if an observed gas ratio value is unlikely to be produced by a normally functioning LTC. The calculated Caution (2) and Warning (3) limits have a statistical significance level of 5% and 1%, respectively.

5.3 Maintenance of LTC DGA norms

Whether or not the limits used for LTC DGA fault detection are statistically derived, it is important to understand that the choice of limits is partly a matter of policy and budget. Presumably if a limit has been set on a gas concentration or gas ratio, there must be some practical consequences—extra attention, more frequent sampling, additional testing, or even an outage—if an LTC exceeds that limit. With that in mind, it is easy to see that the limits, once set, must be reviewed periodically for adequate performance and adjusted if necessary so that the results are consistent with operation and maintenance requirements. Failure to do so is likely to result in economic loss, either through inefficiency and waste in the DGA testing program or through preventable equipment failure.

6. Procedure for interpretation of LTC DGA data

6.1 Types of samples

There are at least four types of LTC DGA samples based on their context and purpose. The interpretation of DGA results from these different types of samples will be different in various obvious respects.

A *baseline* sample is the first sample taken upon initiation of LTC DGA. Testing is also recommended after energization, repair, or oil treatment. A baseline sample is where either there are no previous samples or there is reason to believe that the gas content of the oil measured in previous samples was altered by repair, oil treatment, or other external factors.

Screening samples are taken at regular intervals, typically from six months to one year, for the purpose of checking whether the LTC is showing signs of deterioration or a fault. Screening samples can be compared with one another and with the most recent baseline sample to check for consistency.

Investigative samples are taken at shorter intervals than screening samples for the purpose of confirming Caution (2) or Warning (3) (numbers represent result codes, see 6.3.1). Investigative samples are also useful for following fault development.

A *check* sample is taken soon after another sample to confirm apparently abnormal results or to replace earlier results that may be unusable due to irresolvable data quality problems.

If available, an on-line monitor can be used to provide DGA data results directly.

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6.2 Data quality review

The interpretation of LTC DGA results should begin only after a data quality review to identify and correct data quality problems. It is important to understand that a review of data cannot identify all data quality problems. Therefore, when surprising or alarming results are obtained, it is highly advisable to resample to confirm those results.

Data quality review most often amounts to comparing current test information to prior test information. In the case of an initial sample, the reviewer is limited to reasonableness checks. Current results should always be compared with previously recorded data. In many cases, problems can be resolved by asking the laboratory to check its records. When data quality problems cannot be corrected, resampling is recommended. The following list includes some of the most commonly encountered problems.

— Transcription and typographical errors—Extra digits, digits interchanged, digits dropped, values (especially oxygen and nitrogen) interchanged.

— Missing or duplicated data—Values omitted or values from a previous sample carried forward.

— Misidentified or swapped sample—Incorrect equipment identification, corrupted serial number, and transformer main tank sample mislabeled as LTC sample.

— Cross-contamination—Carryover of gases from one sample to the next during sampling or lab processing, causing sudden or strange changes. Proper use and strict adherence to ASTM D923 and ASTM D3612 should mitigate this as an issue.

— Wildly inconsistent values—Can result from mislabeled samples, sampling problems, measurement problems.

— High nitrogen or oxygen level—For an air-breathing LTC filled with mineral oil, unless the headspace pressure is significantly greater than 1 atmosphere, the concentration of nitrogen should be below 80 000 μL/L (ppm_v) and the concentration of oxygen should be below 40 000 μL/L (ppm_v).

— Low nitrogen level—An extremely low nitrogen concentration in LTC oil is usually a data error, since LTC oil is generally air-saturated when new, and the nitrogen is not consumed.

— Invalid nitrogen/oxygen value—The nitrogen/oxygen ratio should not be calculated or interpreted if either nitrogen or oxygen is below 500 μL/L (ppm_v) or above the maximum possible concentration. The minimum possible valid nitrogen/oxygen value in air-saturated mineral oil is greater than 1.8.

— Inconsistent nitrogen/oxygen values—Unexplained extreme inconsistency of nitrogen/oxygen for a single LTC over a series of samples is a sign of poor data quality.

6.3 Interpretation of DGA data

After resolving any data quality issues, interpretation of the DGA data can be carried out. The products of the interpretation are a numeric result code, a tentative diagnosis (if there is any indication of a fault) or recommended action, and a recommended interval or date for the next sample. If the LTC has multiple oil compartments, the DGA data for each compartment must be interpreted separately, according to its own DGA norms.

6.3.1 Result code

The result code is an indication of relative failure risk or fault severity, based upon comparison of gas concentrations and gas ratios with their respective limits. Suggested result codes are found in Table 1.

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Table 1—LTC DGA result codes

Code	Code description
Normal (1)	No fault is detected
Caution (2)	Weak indication of a fault
Warning (3)	Strong indication of a fault

If none of the combustible gas concentrations exceeds its minimum reliable limit, the result code is Normal (1), no fault is indicated, and the sampling interval is the normal screening interval. If any combustible gas concentration exceeds its minimum reliable limit, the following steps should be performed:

a) Apply the norms to derive the result code.

b) Calculate the gas ratios for which limits are defined, provided that the relevant gases exceed their respective minimum reliable limits.

c) Assign a score to each of these gas ratios:

   1) Normal (1) if no limit is exceeded.

   2) Caution (2) if the caution limit is exceeded

   3) Warning (3) if the warning limit is exceeded

Likewise, compare each combustible gas concentration with its respective caution limit, if defined. Assign a score of Caution (2) to any gas concentration exceeding its caution limit, and assign Normal (1) to gas concentrations that do not exceed their respective caution limits.

The result code for the sample is the maximum of all gas ratio and gas concentration scores.

Fault diagnosis for LTCs based on DGA data alone is not an exact science and needs to be undertaken with caution. An abnormal value of the ethylene/acetylene or TDHG/acetylene ratio usually indicates coking or deterioration of arcing contacts. Gas concentrations exceeding their caution limits—especially if there is more than one gas involved—often indicate abnormal tap-changing activity or some kind of fault. Comparison with case histories of similar LTCs, if available, may provide a means of arriving at a better diagnosis.

For a borderline caution indication, it may suffice to recommend a shortened sampling interval, such as 90 days instead of one year, so that the progress of the fault can be tracked and mitigating action taken at a convenient time. For a more definite fault indication, an immediate check sample should be ordered. If the presence of a serious fault is confirmed, consideration should be given to conducting other kinds of testing, such as infrared scans and internal inspection. More frequent sampling is recommended to detect rapid equipment deterioration while waiting for an opportunity to take corrective action. An on-line DGA monitor could be installed to provide gassing dynamics and an alarm on any condition change.

In air-saturated mineral oil, the nitrogen/oxygen ratio is close to 2. A free-breathing LTC compartment allows sufficient exchange of gases with the atmosphere to compensate for the gradual consumption of oxygen by oil aging and keep the nitrogen/oxygen ratio between 1.9 and 4.5. A ratio value that gradually increases above the typical range, especially if the ratio is 8 or higher, may indicate that the breather port or desiccant breather is not functioning properly. In that case, abnormally high concentrations of hydrogen and other combustible gases may accumulate because diffusion out of the tank is blocked.

The nitrogen/oxygen ratio in a sealed-type LTC compartment with a vacuum/overpressure relief valve is initially close to 2 when the LTC is filled with air-saturated mineral oil. Depending on operation of the relief valve, aging-related oxygen consumption can drive the nitrogen/oxygen ratio higher than would be normal for a free-breathing type. Accumulation of combustible gases may also occur under normal operating conditions. Extremely high nitrogen/oxygen (above 8) with high levels of accumulated combustible gases may indicate that the relief valve is plugged or inoperative.

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See the case histories presented in Annex C for examples of application of this method.

6.3.2 Comparison with similar LTCs

Comparison of DGA data for LTCs of the same type, operating under similar conditions, is very useful. The purpose of such comparison is to identify, graphically or numerically, any LTCs, that might show large unexplained differences from their peers with respect to gas concentrations, gas ratios, or patterns of change over time. This approach can:

— Supplement conclusions based on the application of DGA limits and verify their reasonableness.

— Provide an alternative to limit-based DGA interpretation if limits are unavailable.

6.3.3 Arcing resistive type tap changers

Usually, tap-changers of the arcing type generate huge amounts of arcing gases due to their normal arc-breaking activity. Recent work published by CIGRE [B5] indicates that some arcing resistive type LTC models generate large amounts of heating gas (especially ethylene) instead of acetylene without any fault. This behavior is usually associated with partial load, e.g., it occurs mainly in network applications.

The built-in transition resistors are only stressed during the load switching operation, which is typically performed within 40 ms to 120 ms with the switching time depending on the specific tap changer model. According to IEEE Std 57.131™, the resistors are designed to have a maximum temperature rise of 350 K after multiple operations at 1.5 times rated current. In normal service, the temperature rise will be lower. Generally resistors have been known to produce small amounts of dissolved heating gases.

Some evidence exists that the ethylene/acetylene ratio tends to increase in service with age, sometimes exceeding a value of 200, suggesting either increased heating of LTC parts or, possibly, an increased tendency for the oil to form ethylene. Observations indicate that ethylene may form under certain operating conditions with the occasional and short-lived heating associated with load switching, design, and oil conditions that have not been regarded so far. Studies are ongoing to identify which operating, design, and oil conditions may contribute to the enhanced production of ethylene vs. acetylene.

Therefore, if heating gas concentrations exceed a caution limit or cause any gas ratio to exceed a limit, it is advisable to investigate whether the noticeable heating gas formation can be explained by this exceptional behavior of arcing resistive type LTC types before concluding that a fault is present. Furthermore, limits derived for one population of arcing resistive type LTCs may not be applicable to another population of similar LTCs, as they may experience different load conditions.

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Annex A

(informative)

Generic classification scheme for LTC types

Table A.1 provides a suggested method to categorize LTCs by switch type, tap selector location, and tank breathing design. This grouping may be beneficial to an end user that has an insufficient population of any one manufacturer and type of LTC to establish statistical norms.

Table A.1—Generic LTC type classifications

A	Arcing switch (Arcing contacts are tap-changer contacts where switching arcs occur in oil during a normal tap change operation.)			
V	Vacuum (Vacuum contacts have the switching arc contained in a vacuum bottle.)			
	R[a]	Resistor type (bridging through resistors)		
	X	Reactor type (bridging through a reactance)		
		S	Arcing switches in separate compartment from non-arcing switches	
		A	All oil-immersed switches in one compartment, arcing-type selector	
		N	All oil-immersed switches in one compartment, non-arcing	
			S	Sealed tank (with or without pressure relief)
			B	Non-sealed

[a]Refer to 6.3.3 for cautions regarding resistive type tap changers.

Examples

a) An example classification for a vacuum interrupter type LTC with reactor and separate tap selector in a sealed tank is “VXSS.” The end user could consider grouping all VXSS type LTC compartments to develop DGA norms.

b) An example classification for an arcing type LTC with reactor and all oil-immersed switches, including arcing, in one non-sealed (free breathing) compartment is “AXAB.” The end user could consider grouping all AXAB type LTC compartments to develop DGA norms.

c) An example classification for an arcing type LTC with resistors and all non-arcing oil-immersed switches in one sealed compartment is “ARNS.” The end user could consider grouping all ARNS type LTC compartments to develop DGA norms.

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Annex B

(informative)

Sample procedures for statistical derivation of limits

As suggested in 5.2 of this guide, DGA norms must be tailored for an individual LTC model, or for a generic LTC type as defined in Annex A. For LTCs with dual oil compartments, norms for each oil compartment should be developed separately because of the different amount and intensity of arcing in oil in each compartment. Likewise, data from sealed compartments should not be mixed with data from non-sealed compartments for development of norms. LTC DGA norms may include the following:

— Caution (2) and Warning (3) limits for the ethylene/acetylene ratio (and other ratios that are deemed informative)

— Caution (2) limits for concentrations of methane, ethylene, acetylene, and TDHG

The procedures described here are based on the availability of a DGA data set consisting of one or more DGA samples each for many LTCs of the particular LTC type in question. The limits derived in this way must be used with discretion. In some cases, factors such as severe data quality problems or an inadequate quantity of data may make the statistical derivation of appropriate DGA limits impractical. In such cases, it may be possible to obtain limits for similar LTCs from an analytical laboratory or other entity having a suitable DGA database. If it is not possible to derive or find suitable DGA limits, the approach described in 6.3.2, comparing DGA results for groups of similar LTCs operating under similar conditions, may still be applicable.

Adjustments to all the limits derived following these procedures may be needed to achieve performance appropriate to local operating conditions and equipment. LTC DGA norms should be subjected to periodic performance review and adjusted as required to meet local requirements regarding operation and maintenance.

B.1 Statistical terminology and basis of the derivation procedures

The fault detection method recommended in this guide is based on the proposition that faults or extreme stress in LTCs often cause the formation of unusually high dissolved combustible gas concentrations or unusually high values of certain combustible gas ratios such as ethylene/acetylene. The combustible gas ratio values found in most populations of non-faulty LTCs operating under normal conditions can be described by a lognormal probability distribution. Demonstration of lognormal distribution for LTC DGA ratios makes it possible to derive limits that make fault detection (at a Caution or Warning level) a form of statistical hypothesis testing. When calculating those limits, outliers must be excluded from the gas ratio data for assurance that the limits are based almost entirely on data representing non-faulty, normally operating LTCs.

The distributions of combustible gas concentrations found in populations of non-faulty LTCs operating under normal conditions are often too irregular to fit any probability distribution that is mathematically convenient to work with. Because of that, it is impractical to apply the full statistical hypothesis testing method that is used with LTC gas ratios. Instead, Caution limits for gas concentrations are defined in terms of a statistical outlier limit identifying gas concentrations that are so extreme that they can be suspected to be the result of faults or unusual stress.

The collection of all data points to be used for a statistical analysis is called a *data set*. For example, the collection of all ethylene concentrations available for deriving an ethylene caution limit for a particular type

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of LTC is a data set, and the individual measured ethylene concentrations belonging to that collection are called *data points*. A number derived or calculated from a data set is called a *statistic*. One commonly used statistic is the size N of the data set, which is simply the number of data points in the set. Another is the mean, which is the sum of all the data points in the set, divided by the size of the data set. For some data sets, the standard deviation S is useful [see Equation (B.1)]:

$$S = \sqrt{\frac{\sum_{1}^{N}\left(X_i - M\right)^2}{N-1}} \tag{B.1}$$

where

X_i is a specific data point value
N is the number of data points in the set
M is the mean

By sorting the data set in order of increasing size of its data points, order statistics are obtained. For example, the order statistic $X_{[14]}$ is the value of the fourteenth data point, counting up from the smallest.

The quartiles Q1, Q2, and Q3 of a data set are numbers (not necessarily equal to any data point) that, roughly speaking, divide the data set into quarters. To calculate the first quartile Q1, also known as the 25th percentile, calculate 0.25(N+1) and express it as a whole number k and a fractional part d. Then, as shown in Equation (B.2):

$$\mathrm{Q1} = X_{[\mathrm{k}]} + \mathrm{d}(X_{[\mathrm{k+1}]} - X_{[\mathrm{k}]}) \tag{B.2}$$

where

d is the fractional part of the number calculated as 0.25(N+1)
$X_{[\mathrm{k}]}$ is the value of the kth data point, counting up from the smallest value
$X_{[\mathrm{k+1}]}$ is the value of the $\mathrm{k+1}^{\mathrm{th}}$ data point, counting up from the smallest value

Similarly, the second quartile, Q2, also known as the *median*, is calculated from 0.5(N+1), and the third quartile, Q3, also known as the 75th percentile, is calculated from 0.75(N+1). The interquartile range IQR is defined as shown in Equation (B.3):

$$\mathrm{IQR} = \mathrm{Q3} - \mathrm{Q1} \tag{B.3}$$

The quartiles and interquartile range of a data set can be used to calculate upper outlier limits, as shown in Equation (B.4) and Equation (B.5):

$$\mathrm{U1} = \mathrm{Q3} + 1.5 \cdot \mathrm{IQR} \tag{B.4}$$

$$\mathrm{U2} = \mathrm{Q3} + 3 \cdot \mathrm{IQR} \tag{B.5}$$

For the purposes of this guide it is not necessary to delve into the details of fitting a probability distribution to a data set. It is sufficient to understand that when a data set fits a particular probability distribution, the quartiles and other percentiles of the data set approximate the corresponding quartiles and percentiles of the probability distribution reasonably well so that the probability distribution can be used as a mathematical model of the data set and of future additional data produced in the same way from the same source. This notion of fitting a probability distribution to a data set is the basis for the practice of using percentiles of the probability distribution, instead of percentiles of the data set, as limits for hypothesis testing. In the case of combustible gas ratios from LTC DGA data, the probability distribution is lognormal.

Lognormal distributions are extremely convenient to work with. If a data set fits a lognormal distribution, then the natural logarithms of all the data points form another data set that fits a normal, or Gaussian,

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distribution. Percentiles of that normal distribution can be derived according to familiar formulas as hypothesis testing limits. Those limits can be translated, by means of the exponential function, back into the corresponding percentiles in the original lognormal distribution.

B.2 Derivation of gas concentration caution limits

To derive gas concentration caution limits for a specific model or generic type of LTC, collect the available DGA data for those LTCs, preferably consisting of one or more samples each for 100 or more LTCs. If limits are based on data from fewer than fifty LTCs, they must be used with extra caution and adjusted as required in light of future experience.

For each of the gases methane, ethane, ethylene, and acetylene, calculate the U2 outlier limit according to Equation (B.2), Equation (B.3), and Equation (B.5). Most spreadsheet and statistical software has built-in functions for calculating the quartiles of a data set. Note that there are many variant methods for calculating quartiles from data, so results may differ slightly depending on the particular software used to do the calculations. Such differences are not large enough to affect the validity of the results.

The U2 outlier limit is recommended as a fault detection caution limit instead of the 90th or other percentile because it is a standard statistical criterion for identifying values that seem too extreme to be part of the same population as the majority of the data points.

When tested on data for several common LTC types, including some vacuum type LTCs, this limit was found to flag about 5% to 10% of the samples, depending on the number of questionable LTCs in each group, as unusual. Of course, data percentiles can be used if there is some reason to prefer them.

Before operational use, it is important to examine the derived caution limit for reasonableness and test it against local historical data, if possible, to verify good performance. In some cases, a small data set or a very irregular distribution of gas concentrations can invalidate this procedure and produce an unreasonable result. Comparison with data from known cases in which unusually high gas concentrations were found to be fault related can help in deciding on an appropriate order of magnitude for the caution limit, if there is any doubt. If testing reveals that the statistically derived limit is too strict or too lax relative to local operation and maintenance requirements, the limit value should be adjusted as needed to obtain satisfactory results.

Concentrations of dissolved acetylene and other combustible gas concentrations in vacuum type LTCs are typically at or near the method detection limit. It is possible that the distribution of the gas concentrations in such cases may result in extremely small or values near the method detection limit for statistically derived caution limits. When that happens, caution limits must be chosen on the basis of experience and engineering judgment.

For example, if 75% of the acetylene concentrations for a group of vacuum type LTCs was below the method detection limit, the U2 limit calculated for acetylene would be zero. In that case, the 90th percentile (if not also zero) could be considered or an arbitrary caution limit could be set based on operational experience with similar LTCs, with due regard to operating policies and requirements as well as budgetary constraints on inspections and supplementary testing.

B.3 Derivation of gas ratio limits

The following steps define a procedure for deriving caution and warning limits for a combustible gas ratio. The following example illustrates the process for the ethylene/acetylene ratio. Apply the same process for the other gas ratios of interest.

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To derive gas ratio caution and warning limits for a specific model or generic type of LTC, collect the available DGA data for those LTCs, preferably consisting of one or more samples each for 100 or more LTCs. If limits are based on data from fewer than fifty LTCs, they must be used with extra caution and adjusted as required in light of future experience. If there are too few LTCs or too little data to support good statistical results, consider using default ethylene/acetylene ratio limits of 0.5 (for caution) and 1.0 (for warning) and then adjusting those on the basis of subsequent practical experience to obtain good performance with the given population of LTCs.

Start with all the available DGA samples for the specified model or generic type of LTC.

1) Eliminate from consideration all samples in which either ethylene or acetylene is less than its minimum reliable value.
2) Construct a data set containing the ethylene/acetylene ratio for each remaining sample.
3) Using the preceding Equation (B.2) to Equation (B.5), calculate Q1, Q3, IQR, U1, and U2 for the gas ratios.
4) Eliminate from the data set all ratio values greater than or equal to U2.
5) Construct a new data set consisting of the natural logarithms of the data points remaining in the data set of non-outlier gas ratios.
6) For the data set of logarithms, calculate the size N, the mean M, and the standard deviation S.
7) Calculate the 95th and 99th percentiles of the normal distribution with mean M and standard deviation S, as shown in Equation (B.6) and Equation (B.7):

$$P_{0.95} = M + 1.65*S \tag{B.6}$$

$$P_{0.99} = M + 2.33*S \tag{B.7}$$

8) Calculate the 95th and 99th percentiles of the lognormal distribution representing the non-outlier gas ratios, as shown in Equation (B.8) and Equation (B.9):

$$C_{0.95} = \exp(P_{0.95}) \tag{B.8}$$

$$C_{0.99} = \exp(P_{0.99}) \tag{B.9}$$

The lognormal percentiles $C_{0.95}$ and $C_{0.99}$ are usually appropriate for use as gas ratio caution and warning limits, provided that the final calculations are based on more than 30 logarithm values (N>30). It is preferable to base the limit derivation on 100 or more logarithm values, if at all possible.

Before operational use, it is important to check the derived limits for reasonableness and to verify that they perform well when applied to local historical data. In some cases, a small data set or a very irregular distribution of gas concentrations can cause this statistical procedure to produce unreasonable limits. Ethylene/acetylene and TDHG/acetylene limits for non-vacuum LTCs often fall between about 0.2 and 5, and derived limits outside that range should be checked carefully for good performance with historical data and with documented cases of known faults. Another reasonableness check is that the $C_{0.95}$ and $C_{0.99}$ limits are usually slightly less than the respective outlier rejection limits U1 and U2 recorded in the preceding step 8). Certainly it is very suspicious if $C_{0.95}$ and $C_{0.99}$ are very much larger than U1 and U2. If $C_{0.95}$ and $C_{0.99}$ do not seem reasonable or do not perform well, U1 and U2 may be considered for use as tentative Caution (2) and Warning (3) limits. For small populations of LTCs or small collections of test data, a practical compromise to obtain an appropriate limit is to use the largest non-outlier, i.e., the largest observed gas concentration that is not greater than U2, as the caution limit.

The $C_{0.95}$ and $C_{0.99}$ limits calculated according to this procedure represent critical values for a statistical hypothesis test with significance level 0.05 or 0.01. The test, applied to the ethylene/acetylene ratio from a new DGA sample for an LTC in the chosen population, has the following null hypothesis:

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H0: The gas ratio was randomly chosen from the lognormal distribution of non-faulty LTCs operating under normal conditions.

The alternative hypothesis of the test, which is accepted if the gas ratio exceeds either of the limits, is:

H1: The gas ratio is too large to have been randomly chosen from the lognormal distribution of non-faulty LTCs operating under normal conditions.

An important advantage of deriving the gas ratio limits in this way as critical values for a statistical hypothesis test is that the approximate probability of a "false positive" result (gas ratio exceeds the caution or warning limit but the LTC is neither faulty nor reacting to unusual stress) is known and can be chosen. If, for example, a significance level of 0.10 is preferred instead of 0.05 for the caution limit, the normal distribution z-value $z_{0.90} = 1.28$ can be used instead of the factor $z_{0.95} = 1.65$ used in Equation (B.5).

Typical values of acetylene and other combustible gas concentrations in most vacuum type LTCs are not detected or very close to zero. It is possible that most of the gas concentration values large enough for calculating ratios may already exceed the caution limits, so consequently there is no basis for the derivation of ratio limits. In such cases, fault detection should be based on gas concentration limits alone.

B.4 Examples of LTC DGA limit derivations

The methods previously described for deriving gas concentration and gas ratio limits are demonstrated here by examples based on 726 DGA samples from a single LTC model (arcing in oil, single compartment, reactive) at one large electric utility. The corresponding LTC type from Annex A is AXNB.

A spreadsheet tool for carrying out these calculations is available at the following URL: http://standards.ieee.org/downloads/C57/C57.139-2015/IEEE-Std-C57_139-2015-Calculator-D1.xls. It is initially loaded with the DGA data for this example.

B.4.1 Caution limit for ethylene

Following the recommendations of B.2, all of the ethylene concentrations from the 726 DGA samples are collected as a single data set. The quartiles, IQR, and U2 outlier limit are calculated, producing the following results:

Q1 = 3.0, Q3 = 14.0, IQR = 11.0, and U2 = 47.0

The proposed ethylene caution limit for this population of LTCs is 47.0 µL/L (ppm_v).

Of the 726 ethylene concentrations, 54 (about 7%) exceed 47.0. That does not seem unreasonable. If, in practice, 47.0 as a Caution (2) limit for ethylene causes too many LTCs to be flagged for extra attention, the limit could simply be raised enough to remedy that problem.

B.4.2 Caution and warning limits for ethylene/acetylene ratio

In this example, the Caution (2) and Warning (3) limit derivation procedure in B.3 is illustrated for the ethylene/acetylene gas ratio. The procedure is the same for all other combustible gases ratios for which Caution (2) and Warning (3) limits are desired.

1) From the original 726 DGA samples, eliminate from consideration all the samples where the ethylene concentration or the acetylene concentration is less than the minimum reliable limit of 10 µL/L (ppm_v), leaving 253 samples.

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2) From those remaining 253 DGA samples, create a data set of 253 ethylene/acetylene ratio values.

3) Using the preceding Equation (B.2) to Equation (B.5), calculate Q1, Q3, IQR, U1, and U2 for the gas ratios, obtaining:

   Q1 = 0.155, Q3 = 0.310, IQR = 0.155, and U2 = 0.777

4) Eliminate from the data set all ratio values greater than or equal to U2, leaving 229 non-outlier data points.

5) Construct a new data set consisting of the natural logarithms of the data points remaining in the data set of non-outlier gas ratios.

6) For the data set of logarithms, calculate the size N, the mean M, and the standard deviation S, obtaining:

   N = 229, M = -1.596, and S = 0.471

7) Calculate the 95th and 99th percentiles of the normal distribution with mean M and standard deviation S:

   $P_{0.95} = M + 1.65S = -1.596 + (1.65)(0.471) = -0.81885$

   $P_{0.99} = M + 2.33S = -1.596 + (2.33)(0.471) = -0.49857$

8) Calculate the 95th and 99th percentiles of the lognormal distribution representing the non-outlier gas ratios:

   $C_{0.95} = \exp(P_{0.95}) = \exp(-0.81885) = 0.44$

   $C_{0.99} = \exp(P_{0.99}) = \exp(-0.49857) = 0.61$

As a reasonableness check, compare the derived limits to the entire original set of 253 ratio values created in the preceding step 2). Of the 253 original ratio values, 35 exceed the caution limit 0.44. This represents about 5% of the whole collection of 726 samples. Similarly, 31 of 253 exceed the warning limit of 0.61, representing about 4% of all the DGA samples. These are typical of the proportions expected in electric utility LTC DGA data.

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Annex C

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Case histories

C.1 Norms used for Examples 1 and 2

The statistical norms shown in Table C.1 were derived from 645 DGA samples for a common model of LTC (classification type AXNB) at a large U.S. electric utility. They are used in Examples 1 and 2.

Table C.1—Derived statistical norms for Example 1 and 2 LTCs

Variable	Caution (2)	Warning (3)
Ethylene/Acetylene	0.63	0.87
TDHG/Acetylene	1.51	2.33
Methane (CH_4)	237 μL/L (ppm_v)	
Ethylene (C_2H_4)	653 μL/L (ppm_v)	
Acetylene (C_2H_2)	1773 μL/L (ppm_v)	
Heating gas (TDHG)	1213 μL/L (ppm_v)	

C.1.1 Example 1—Badly coked arcing contacts

The Example 1 LTC had very low gas concentrations—so low that it was not meaningful to calculate the gas ratios—from 1999 through 2005 as shown in Table C.2.

Table C.2—Example 1 gas concentrations

Date	H_2 μL/L (ppm_v)	CH_4 μL/L (ppm_v)	C_2H_6 μL/L (ppm_v)	C_2H_4 μL/L (ppm_v)		CO μL/L (ppm_v)	CO_2 μL/L (ppm_v)	TDHG μL/L (ppm_v)	C_2H_4/C_2H_2	TDHG/ C_2H_2
1999-05-02	25	2	0	3	5	97	714	5		
2000-01-12	21	1	1	2	9	37	507	4		
2001-08-16	60	4	2	5	11	213	1333	11		
2002-07-17	56	3	1	8	18	252	47	12		
2003-01-13	0	3	0	2	5	49	582	5		
2004-04-02	0	1	0	1	2	24	303	2		
2005-05-09	41	1	0	2	3	204	613	3		
2006-03-08	0	7	10	42	10	121	903	59	**4.20**	**5.90**

In March 2006, the ethylene, TDHG, and acetylene concentrations were still below their respective caution limits, but high enough to allow the calculation of ratios. Both the ethylene/acetylene and TDHG/acetylene ratios had extremely high values, exceeding their respective Warning (3) limits by a large margin, suggesting severe coking or degradation of the arcing contacts. In October 2006, the fault was discovered by non-DGA means, and internal inspection revealed that the contacts were badly coked.

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C.1.2 Example 2—LTC unable to operate

From 1998 through 2004, the Example 2 LTC at the same electric utility had substantial concentrations of combustible gases shown in Table C.3, although neither the gas concentrations nor their ratios exceeded the statistical norms shown in Table C.1.

Table C.3—Example 2 gas concentrations

Date	H_2 µL/L (ppm_v)	CH_4 µL/L (ppm_v)	C_2H_6 µL/L (ppm_v)	C_2H_4 µL/L (ppm_v)	C_2H_2 µL/L (ppm_v)	CO µL/L (ppm_v)	CO_2 µL/L (ppm_v)	TDHG µL/L (ppm_v)	C_2H_4/C_2H_2	TDHG/ C_2H_2
1998-10-01	23	16	7	84	403	10	508	107	0.21	0.27
1999-05-05	58	41	16	190	1031	16	511	247	0.18	0.24
2000-01-06	54	38	18	203	1052	3	510	259	0.19	0.25
2002-09-03	51	28	17	168	720	126	28	213	0.23	0.30
2003-01-20	85	62	10	145	750	28	494	217	0.19	0.29
2004-01-15	121	66	24	314	1400	27	391	404	0.22	0.29
2005-11-14	146	154	79	845	3300	29	524	1078	0.26	0.33
2006-01-17	235	127	81	849	3500	15	319	1057	0.24	0.30

In November 2005 it was evident that combustible gas concentrations were much higher than in previous samples; in fact, the ethylene and acetylene exceeded their respective caution limits, and TDHG was close to its limit. The gas ratios remained below their caution limits. A sample two months later, in January 2006, confirmed the high gas concentrations. In this situation, investigating why the gas concentrations have increased so much may be considered, even though the ratios do not indicate any problem with the arcing contacts.

Due to the high gas concentrations, the crew looked at the operation counts for the LTC. Operation counts in 2005 (31 907) and 2006 (31 133) were extremely high compared to 8 157 operations in 2004. A maintenance crew discovered in late 2006 that the LTC had a faulty balance beam control.

C.2 Example 3—Burned reversing switch

The statistical norms shown in Table C.4 were derived from 1272 DGA samples from another common type of LTC at several U.S. electric utilities and are used in this example (classification type AXAB).

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Table C.4—Derived statistical norms used for Example 3 LTCs

Variable	Caution (2)	Warning (3)
Ethylene/acetylene	0.42	0.57
TDHG/acetylene	0.63	0.87
Methane (CH_4)	28 μL/L (ppm_v)	
Ethylene (C_2H_4)	59 μL/L (ppm_v)	
Acetylene (C_2H_2)	292 μL/L (ppm_v)	
Heating gas (TDHG)	96 μL/L (ppm_v)	

Dissolved gas analysis indicated an abnormal gassing source inside the LTC. In late 2004, ethylene, acetylene, and TDHG seemed higher, and the ethylene/acetylene and TDHG/acetylene ratios had increased slightly, although both were within their normal range. See Table C.5.

Table C.5—Example 3 gas concentrations

Date	H_2 μL/L (ppm_v)	CH_4 μL/L (ppm_v)	C_2H_6 μL/L (ppm_v)	C_2H_4 μL/L (ppm_v)	C_2H_2 μL/L (ppm_v)	CO μL/L (ppm_v)	CO_2 μL/L (ppm_v)	TDHG μL/L (ppm_v)	C_2H_4/C_2H_2	TDHG/ C_2H_2
2003-04-22	23	9	3	38	145	9	639	50	0.26	0.34
2004-11-09	11	11	5	53	163	37	702	69	0.33	0.42
2006-04-25	45	139	302	1390	77	60	973	1831	18.05	23.78
2006-06-20	5.9	32	204	776	66	63	888	1012	11.76	15.33
2007-05-07	2	1	57	129	5	49	576	187	--	--

By April 2006 there had been a large increase in hot metal gases, putting methane and ethylene above their caution limits. The ethylene/acetylene and TDHG/acetylene ratios had gone far above their Warning (3) limits, suggesting severe coking or deterioration of arcing contacts. These results were confirmed by another sample two months later.

An internal inspection of the LTC found a burned reversing switch. To prevent LTC operation and stop the gassing, the LTC was locked down in the neutral position. By May 2007, almost a year later, gas levels had declined considerably. Acetylene, at 5 μL/L (ppm_v), was extremely low, removing gas ratios from consideration.

C.3 Example 4—Burned contact in vacuum type LTC

Due to the very low combustible gas concentrations that are typically present in vacuum type LTCs, it is not unusual to find that the recommended procedure for deriving statistical norms can produce only a few gas concentration caution limits, and no gas ratio limits. For this example, the statistical procedure was applied to 67 DGA samples for a common type of vacuum type LTC at a large U.S. electric utility (classification type VXAB). This procedure came up with only two combustible gas caution limits: ethylene 32 μL/L (ppm_v) and TDHG 39 μL/L (ppm_v).

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In April 2007, the gas concentrations for one of the vacuum LTCs had extremely low values, which would be expected for this kind of LTC. In May 2008, the gas concentrations were so extremely high that no limits were needed to see that something was wrong. The check sample one month later, in June 2008, confirmed these results. See Table C.6.

Table C.6—Example 4 gas concentrations

Date	H_2 µL/L (ppm_v)	CH_4 µL/L (ppm_v)	C_2H_6 µL/L (ppm_v)	C_2H_4 µL/L (ppm_v)	C_2H_2 µL/L (ppm_v)	TDHG µL/L (ppm_v)	CO µL/L (ppm_v)
2007-04-16	2	1	0	5	0	6	41
2008-05-20	8037	53747	64064	107487	969	226267	35
2008-06-10	7425	52779	76848	122821	990	253438	37
2008-09-11	56	98	86	496	0	680	73
2008-10-10	82	109	90	528	0	727	101

An internal inspection of the LTC found a burned phase B contact. The damaged contact and the insulating fluid were replaced. The sharply reduced but still considerable levels of combustible gas found in the September and October 2008 samples may represent residual gas, but further surveillance is required for assurance that the fault is gone.

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Annex D

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Application of the Duval Triangle 2 to DGA in LTCs

D.1 Introduction

The Duval Triangle 2 for LTCs (see Duval [B2], [B3]) is a graphical representation based on the concentrations of methane, ethylene, and acetylene. Each DGA sample can be plotted on the triangle as a dot whose location is determined by the relative proportions of those three gases. The triangle is used to:

— Distinguish between normal and abnormal LTC gas formation.

— Visualize the evolution of gas formation in an LTC over time.

— View and compare the patterns of gas formation by groups or populations of LTCs.

The triangle is subdivided into zones that are related to common types of conditions seen in LTCs. See Table D.1 and Figure D.1.

Table D.1—Triangle zones

N	Normal operation of an LTC in good condition
D1	Abnormal D1 arcing
X1	Mild overheating
X3	Coking in progress or abnormally strong (D2) arcing
T2, T3	Moderate or severe heating, often related to coking

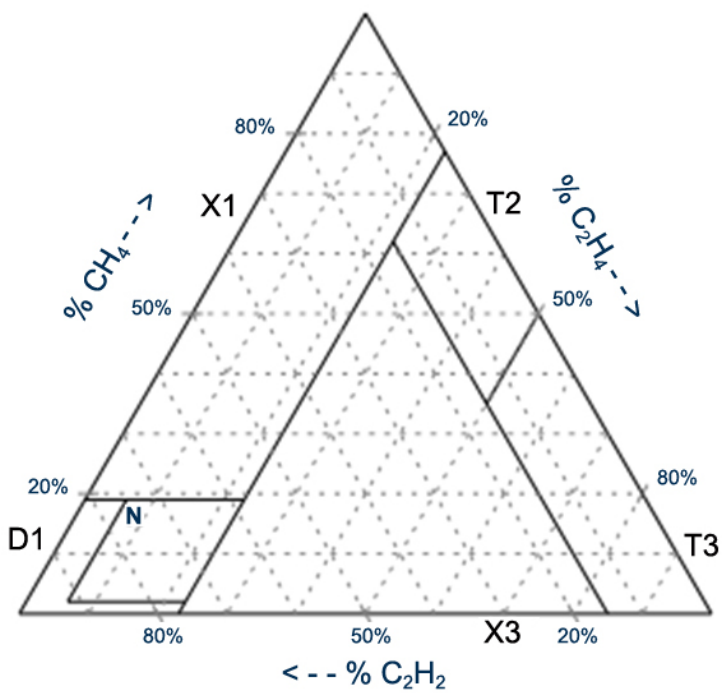

Figure D.1—Duval Triangle 2 for LTCs

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D.2 Data plotting example

An LTC oil sample with 40 ppm_v of methane, 60 ppm_v of ethylene, and 300 ppm_v of acetylene is plotted on the triangle as follows. First, calculate the sum 40 + 60 + 300 = 400 of the concentrations of methane, ethylene, and acetylene. Then for each of the gases calculate its percentage of the total:

— Methane: 40/400 = 10%

— Ethylene: 60/400 = 15%

— Acetylene: 300/400 = 75%

Plot the point of intersection of three lines: the horizontal line to the right from 10% on the methane (left) scale; the line slanting down and to the left from 15% on the ethylene (right) scale; and the line slanting up and to the left from 75% on the acetylene (bottom) scale, as shown in Figure D.2, where in this case the dot is seen to fall within the triangle's N zone.

Note that if any of the three gas concentrations is less than about 10 times its method detection limit, the usefulness of the data for fault identification is doubtful, and the location of the point on the triangle may be somewhat vague due to the effects of measurement uncertainty.

By plotting the data from a series of samples on the same triangle, it is easy to see whether the LTC condition is stable or changing in a gradual or abrupt way. A commonly seen pattern is a cluster of dots in the N zone, representing a long period of normal operation, with the most recent samples trending off to the right into the X3 zone. Usually that pattern indicates progressive coking of arcing contacts.

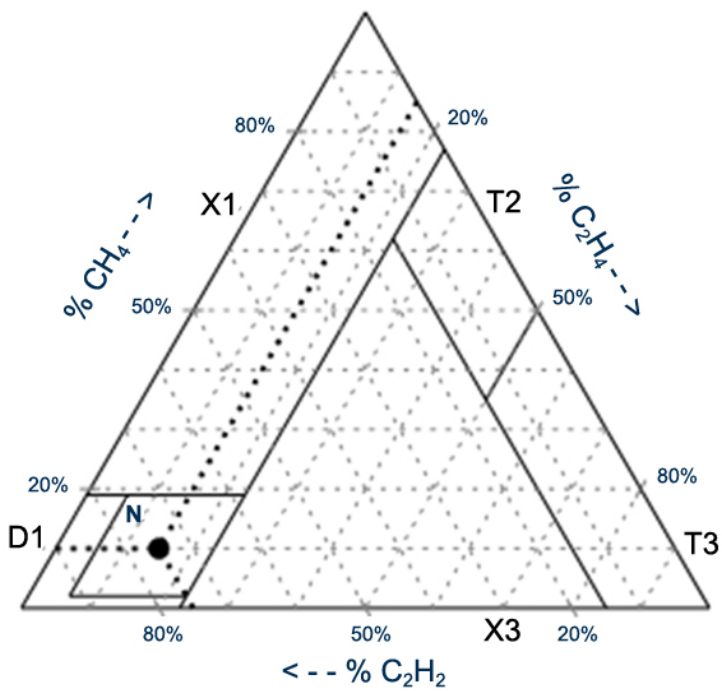

Figure D.2—Plotting data on the triangle

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D.3 Fault evolution example

The 2003 through 2006 data from Example 3 (burned reversing switch, C.2) are plotted in Figure D.3, connected with a thin solid line to show the sequence. There is a large displacement of the sample points from the N zone when the reversing switch problem begins to be apparent in the 2006 DGA data.

Also shown in this figure is a thick line from the apex of the triangle to the base, representing the ethylene/acetylene ratio caution limit of 0.42, taken from Table C.4. If all of the DGA data from the entire population of LTCs from which the limit was derived were plotted on this same chart, approximately 95% of the dots would fall to the left of that limit line.

To plot a line representing a constant value *r* of the ethylene/acetylene ratio, draw a line from the apex of the triangle (where ethylene = 0%) to the base (acetylene axis) at the point where the percent acetylene equals 100 / (*r*+1). In this example, the line terminates at 100 / 1.42 = 70.4 on the %C_2H_2 axis. The same principle applies for drawing lines representing constant values of methane/ethylene (left corner to right side) and acetylene/methane (right corner to left side).

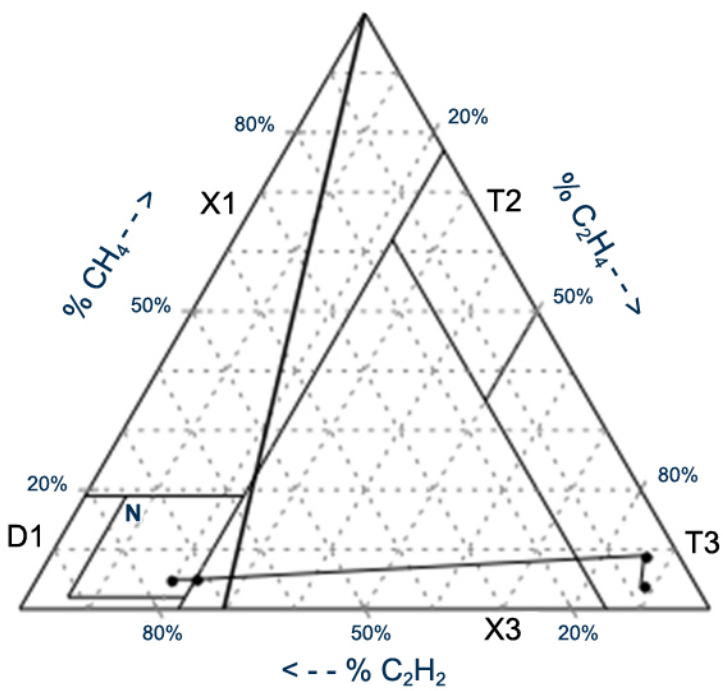

Figure D.3—Burned reversing switch example

D.4 Exceptional LTC types

Some arcing resistive type LTCs in good condition, especially of the in-tank type (classification ARSB and ARAB), can generate gases under normal operating conditions in proportions that in other LTCs would indicate a severe problem. The reason for this behavior is not clearly understood; since the transition resistors are designed not to reach extreme temperatures during tap switching operations (refer to 6.3.3).

Although the usual principle—that dominant ethylene in an LTC is an indication of trouble—is false for these exceptional LTCs, it is still possible to observe what gassing patterns are found during normal operation in fault-free LTCs and then investigate when the gases do not conform to whatever the normal patterns may be. Such exceptional "normal" patterns for several models of resistive type (vacuum or non-vacuum type) LTCs can be visualized as alternative N zones (in addition to the original one) superimposed

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on Triangle 2, depending on the LTC model. For each LTC model exhibiting this exceptional gas formation, the boundaries of the extra N zone(s) must be determined empirically or obtained from the manufacturer.

Figure D.4 is Triangle 2 with four extra normal zones superimposed to show the variety of cases that have been observed. A fifth extra N zone, N2, is not shown because it overlaps N3. Each of these five models of resistive type LTCs, when fault-free, operates in the original N zone or, under certain conditions, in its own extra N zone. Combinations of gas concentrations falling outside N and the LTC model's extra N zone represent unusual behavior or unusual operating conditions and may require investigation. The normal zone(s) of operation of LTCs of the in-tank types, in terms of gassing, should be precisely determined first by users, depending on the type, model, and operating conditions of the LTCs. See Table D.2 for the triangle zone and extra N zone boundaries.

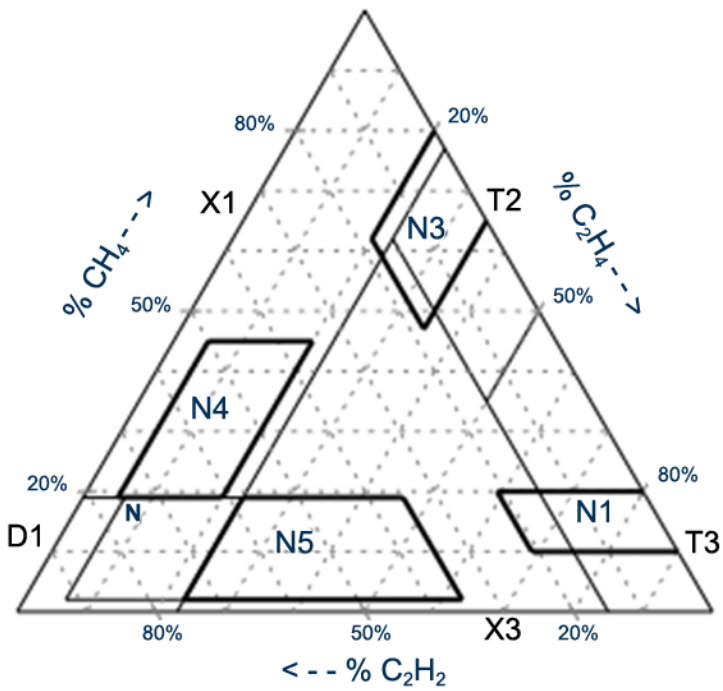

Figure D.4—Extra N zones for four LTC types

— Zone N: most AX** models, many AR** and VR** models; also VXA* models with arcing selector or bypass switches

— Zone N1: some AR** models

— Zone N3 (including the overlapping Zone N2): some VR** models

— Zone N4, N5: some AR**, VR** models

NOTE 1—The character "*" is a wildcard to cover multiple models.

NOTE 2—For practical use, the exact LTC model and the individual operating conditions must be identified.

NOTE 3—Typical examples of LTC models operating normally in zones N to N5 of Figure D.1 and Figure D.4 are indicated in Duval [B4].

NOTE 4—VX** models without arcing selector or bypass switches should rather use the transformer triangle (Duval Triangle 1), since gas formation in them is always abnormal.

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Table D.2—Zone boundaries in Triangle 2

Zone	Boundaries in %
N	CH_4=19, C_2H_4=23, CH_4=2, C_2H_4=6
T3	C_2H_4=50, C_2H_2=15
X3	C_2H_4=23, C_2H_2=15
T2	C_2H_4=23, C_2H_2=15, C_2H_4=50
D1	CH_4=19, C_2H_4=6, CH_4=2, C_2H_4=23
X1	CH_4=19, C_2H_4=23
	Extra N zones for exceptional LTC types
N1	CH_4=20, CH_4=10, C_2H_2=21
N2	CH_4=63, CH_4=44, C_2H_4=28, C_2H_4=45, C_2H_2=20
N3	C_2H_4=20, C_2H_4=35, C_2H_2=18
N4	CH_4=45, CH_4=19, C_2H_4=5, C_2H_4=20
N5	CH_4=19, CH_4=2, C_2H_4=23, C_2H_2=35

D.5 LTC population example

It is sometimes useful to plot large numbers of sample points on a single triangle to investigate the characteristic gas formation patterns of an entire population of comparable LTCs. In Figure D.5 it is apparent that the LTC population has two main modes of deterioration, visible as two distinct streams of dots going away from the N zone to the right. One follows the X1 zone upwards, and the other proceeds horizontally across to the T3 zone. There may also be a third trend from the N zone across X3 to the T2-T3 boundary. Internal inspections might reveal what is different about these modes of deterioration.

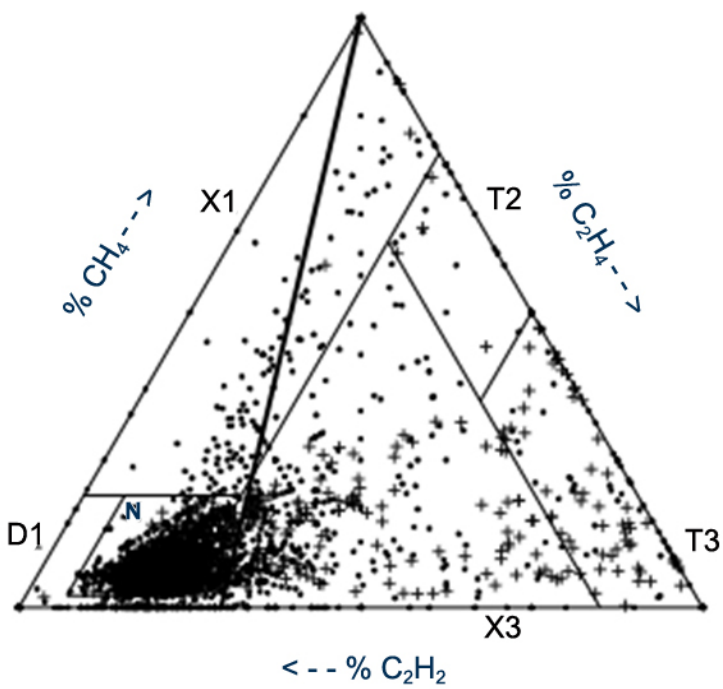

Figure D.5—DGA data for entire LTC population

By color coding the sample dots or using a distinctive symbol to highlight the samples with abnormally high gas concentrations, it is possible to investigate the relationship between high gas concentrations and problems that may be detected with the triangle or with gas ratios. In this figure, the statistical caution limit (0.42) for the ethylene/acetylene ratio is plotted as a dark line from the triangle apex to the base. Samples in which methane, ethylene, or acetylene exceeds its caution limit are plotted as a plus sign.

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Annex E

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Gas ratio nomograms for DGA in LTCs

E.1 Introduction

A *nomogram* is a collection of numerical scales arranged so that a specialized calculation can be performed graphically by plotting a straight line across the scales. Before the advent of electronic calculators and portable computers, nomograms were extensively used for navigation, artillery, photography, and many other engineering and scientific applications. For simplicity, graphical visualization, and availability when electricity and telecommunications are unavailable, preprinted nomogram templates are very useful. Nomograms for quick evaluation of gas ratios important for LTC DGA are presented in this annex.

Each gas ratio nomogram has three numeric scales, representing the numerator gas concentration, the denominator gas concentration, and the gas ratio. The gas concentration scales deliberately omit extra-low gas concentrations that should not be used for calculating gas ratios. A straight line drawn through the three scales, intersecting the two gas concentration scales at the respective observed gas concentration values, intersects the gas ratio scale at the computed value of the gas ratio. It is possible to plot multiple results on a single nomogram for visualizing fault evolution in a single LTC or for comparison of results from different LTCs within the same group.

The nomogram scales can be marked up to increase their usefulness for data interpretation in any particular case. The gas ratio scale can be highlighted or annotated to indicate *caution* and *warning* ranges. For an exceptional LTC type as discussed in Annex D, a suitable extra *normal* range can also be designated on the gas ratio scale. Similarly, the gas concentration scales can be marked up to show their respective normal and abnormal ranges. When a line representing oil sample data is drawn across the annotated nomogram, the statuses of the gas concentrations and their ratio are all simultaneously visible in the diagram.

E.1.1 Heating gas / acetylene ratios

As discussed in 5.2.3, ratios of heating gas to acetylene concentration are useful for detecting abnormal heating, especially heating due to coking, deterioration, or misalignment of oil-immersed arcing contacts. The most useful gas ratios of this kind are ethylene / acetylene and TDHG / acetylene.

The very high March 2006 ethylene / acetylene gas ratio value of 4.2 from C.1.1 Example 1 (badly coked arcing contacts) is plotted on the ethylene / acetylene nomogram in Figure E.1.

Figure E.3 is a heating gas / acetylene nomogram template. The heating gas can be ethylene, TDHG, methane, or ethane.

E.2 Nitrogen / oxygen ratio

A gas ratio nomogram is useful for visualizing and assessing the N_2/O_2 ratio. That ratio can indicate problems with the LTC desiccant breather or vacuum/pressure relief valve, as discussed in 6.3. As explained in 6.2, certain oxygen and nitrogen concentrations and N_2/O_2 ratio values can represent bad data.

In Figure E.2, the N_2/O_2 nomogram has a thin solid line plotted across it connecting N_2 = 70462 with O_2 = 25019 and extending over to N_2/O_2 = 2.8, a ratio value suggesting that the LTC breather is functioning normally. The dashed line plotted on the nomogram connects N_2 = 71922 with O_2 = 8683, extending over to 8.3 on the N_2/O_2 scale, a high value suggesting that the LTC's desiccant breather may need attention.

Figure E.4 is a nitrogen / oxygen gas ratio nomogram template.

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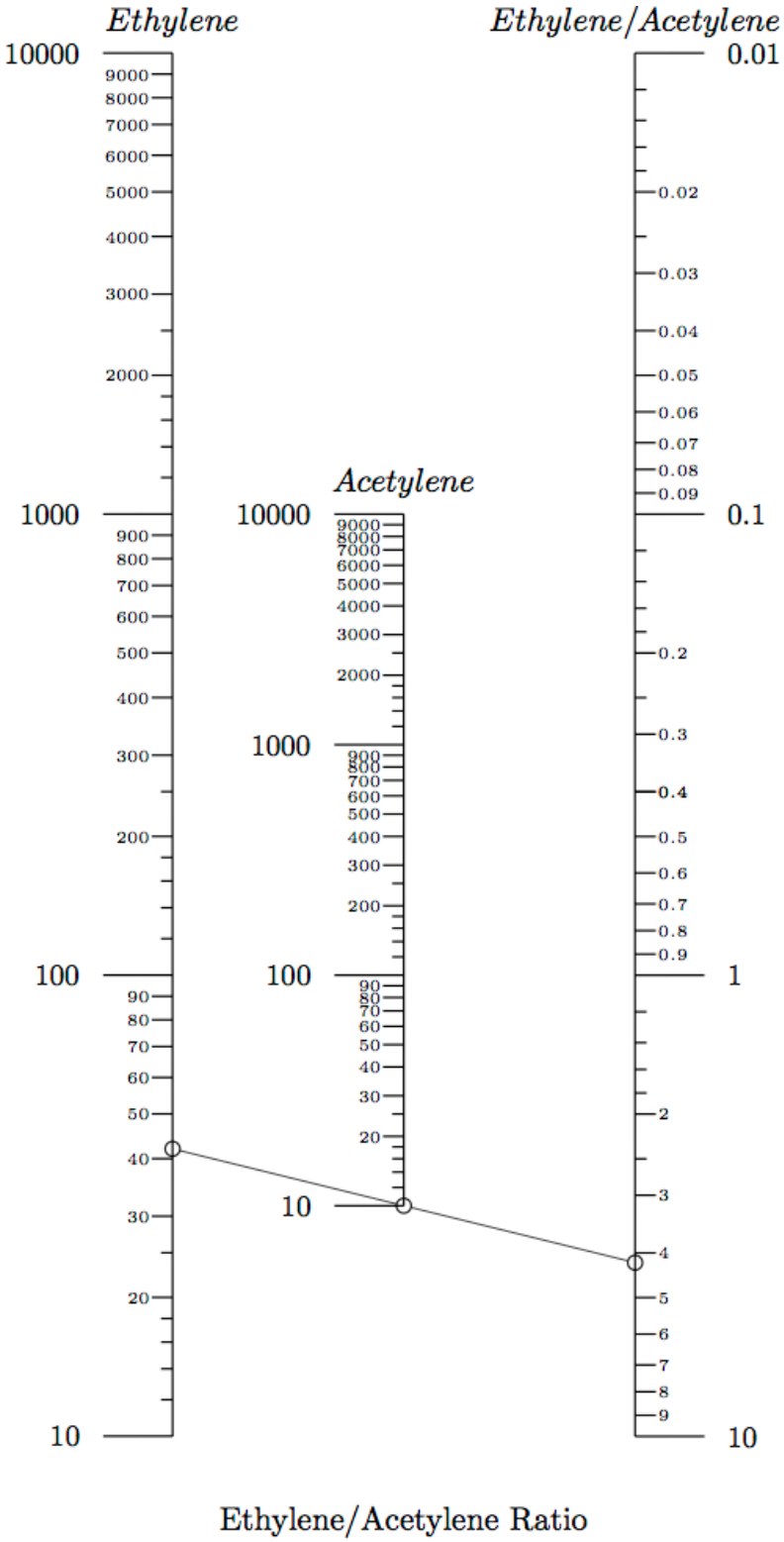

Figure E.1—Ethylene / acetylene gas ratio example

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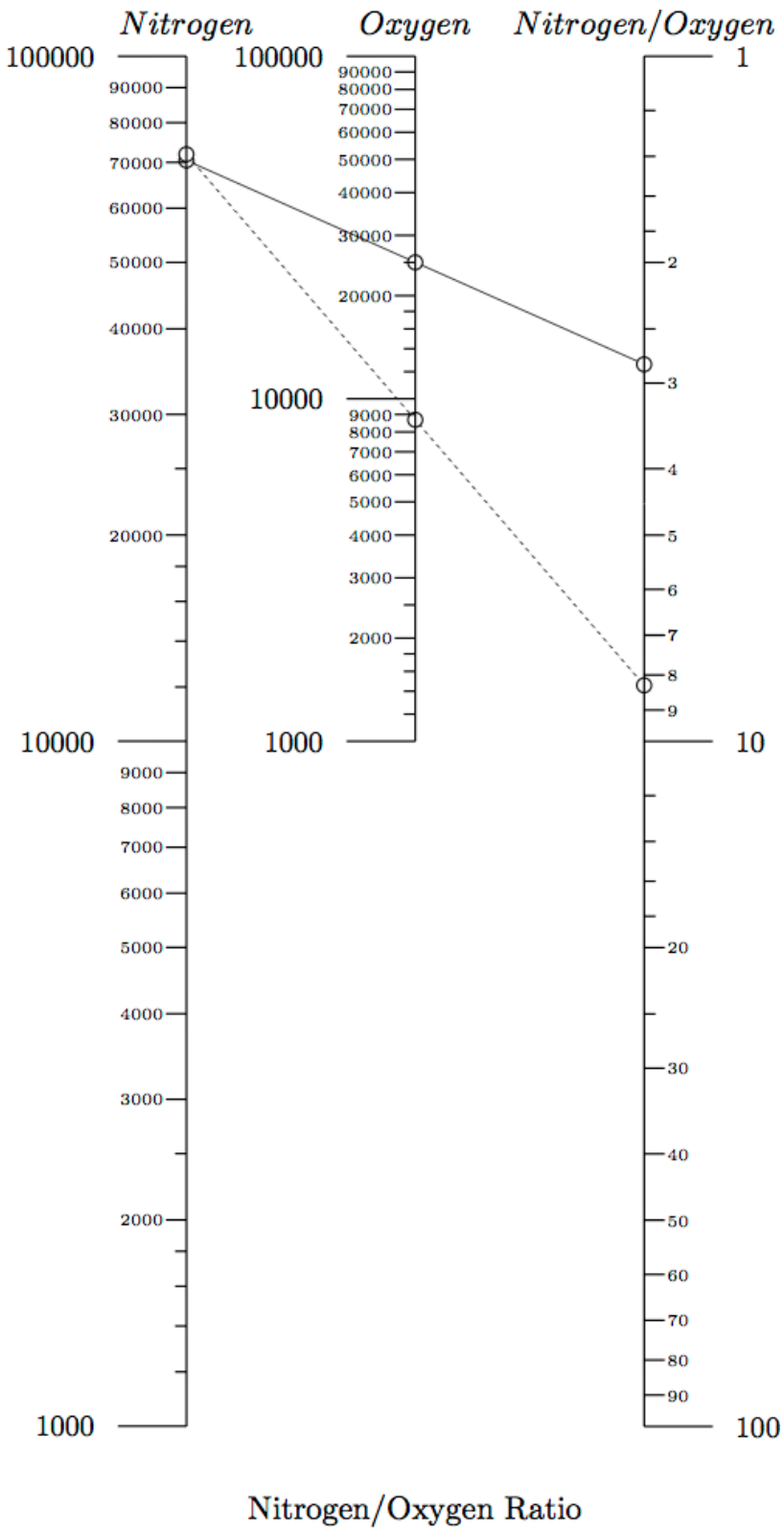

Figure E.2—Nitrogen / oxygen ratio example

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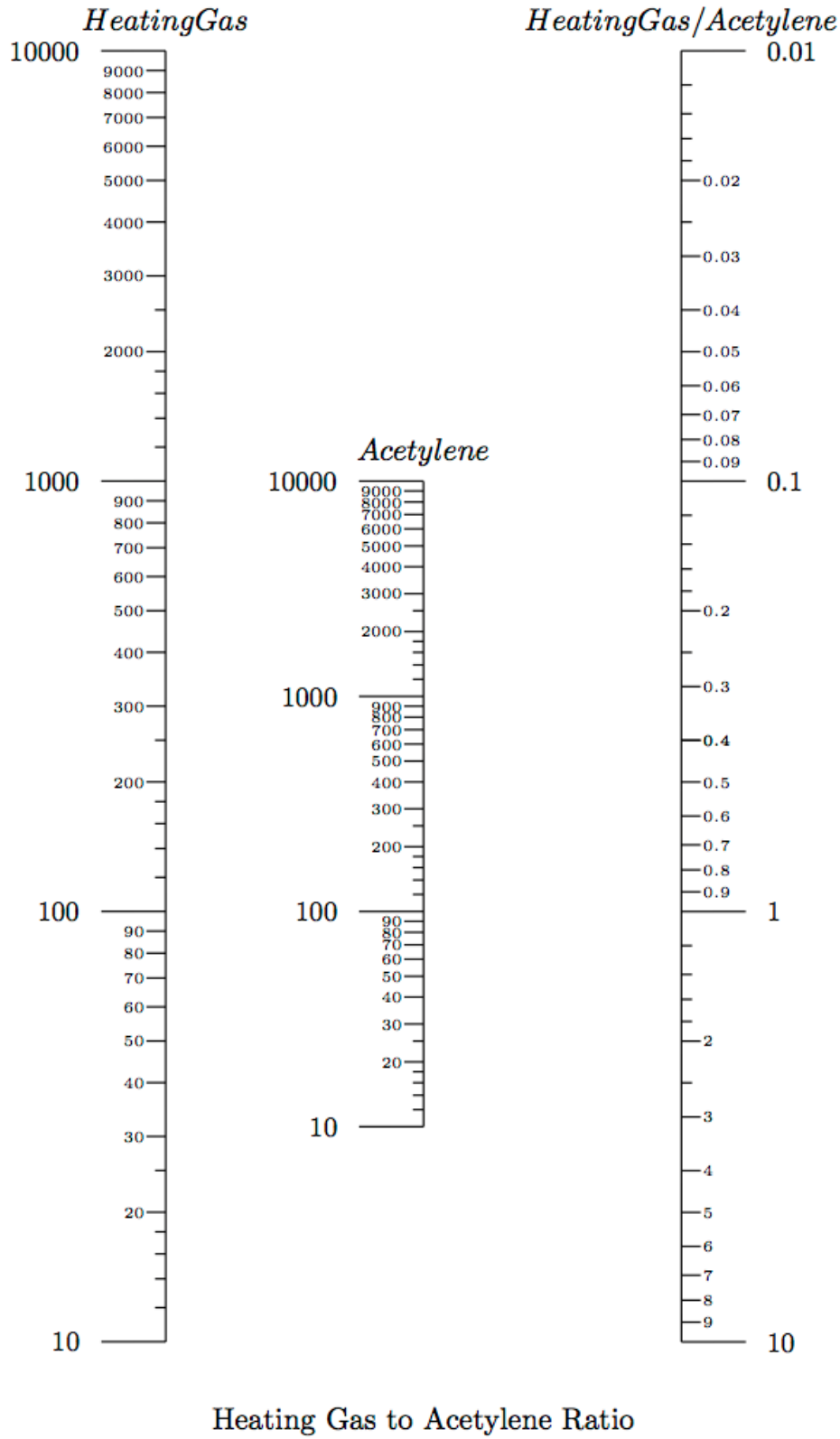

Figure E.3—Heating gas / acetylene nomogram template

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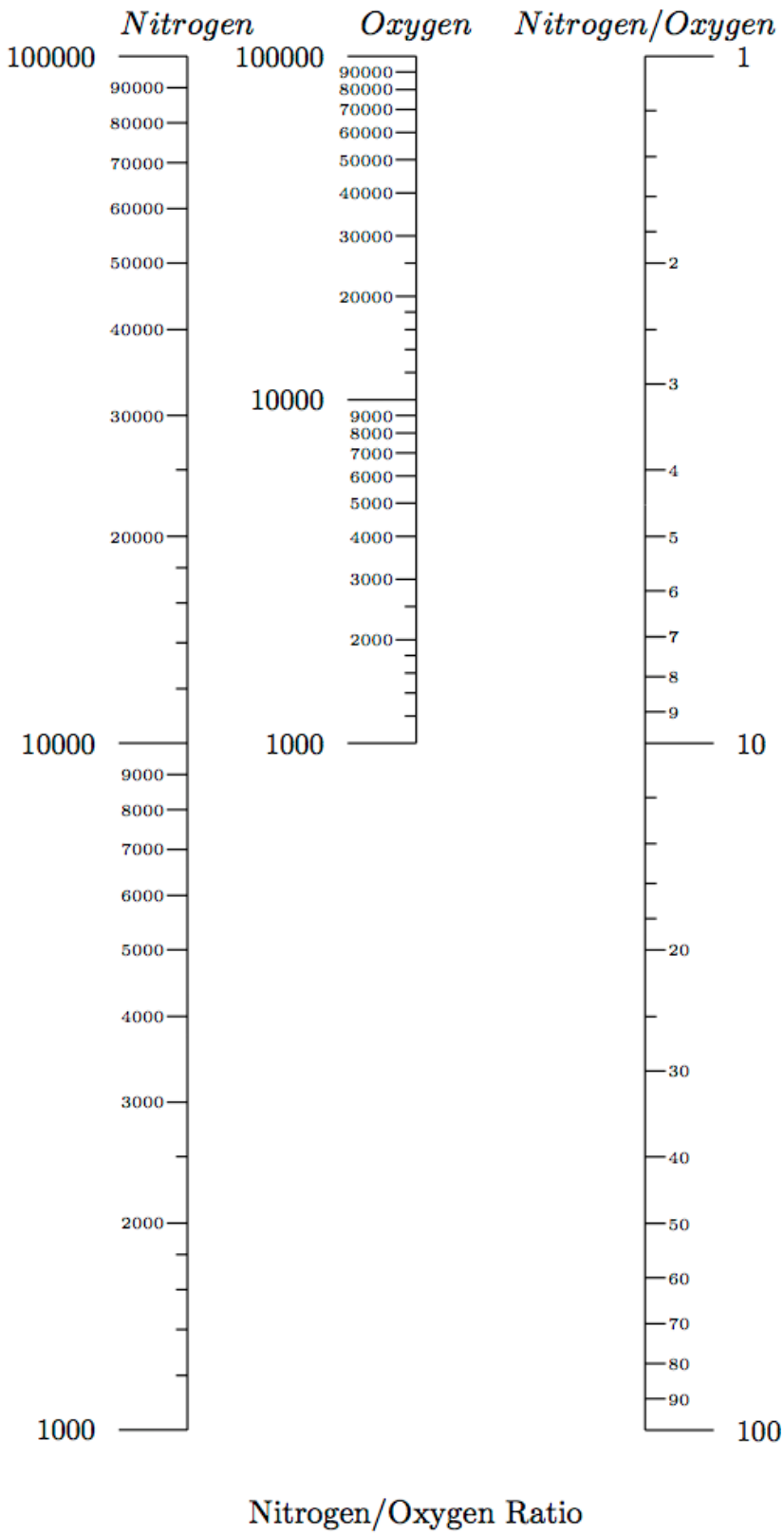

Figure E.4—Nitrogen / oxygen nomogram template

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Annex F

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Bibliographical references are resources that provide additional or helpful material but do not need to be understood or used to implement this standard. Reference to these resources is made for informational use only.

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